

MIGUEL MAURICIO MENDOZA MENDOZA

Zacatecas, Zacatecas, Mexico

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Biomedical engineer with expertise in data analysis, artificial intelligence, and medical equipment management. Focused on scientific research, solving problems through machine learning, developing, AI-based predictive models, and statistical analysis to improve healthcare services. Skilled in medical equipment maintenance, implementing preventive and predictive strategies.

Education

Master of Science in Information Processing

Universidad Autónoma de Zacatecas | August 2023 - August 2025

Bachelor's degree in biomedical engineering

Universidad Autónoma de Chihuahua | August 2015 - December 2020

Professional Experience

Data Engineer & AI Engineer – HeyDonto API

Guadalajara, Mexico | July 2025 – February 2026

- Backend development in Python for transformation, normalization, and standardization of clinical data to the FHIR HL7 standard, enabling interoperability between dental and veterinary PMS systems.
- Design and implementation of a Retrieval-Augmented Classification (RAC) architecture over vector databases, enabling hybrid search (lexical + semantic) to improve integration quality with machine learning registered MLflow models.
- Integration of LLM-based agents using LangGraph (ReAct) to automate data ingestion, classification, and enrichment processes.
- Initial implementation of Knowledge Graphs (KG) to model clinical relationships and enhance information retrieval based on facts and contextual reasoning.

Research Assistant (Data Science) at Universidad Autónoma de Zacatecas

Zacatecas, Mexico | August 2023 – August 2025

- [Interpretable deep learning for breast lesion classification: A SHAP-based comparison of CESM and digital mammography](#)
- [On-device conversational agent for psycho-oncology based on acceptance and commitment therapy manuals](#)
- [Assessing ResNeXt and RegNet Models for Diabetic Retinopathy Classification: A Comprehensive Comparative Study](#)
- [Developed a predictive model for ovarian cancer detection using machine learning techniques and genetic algorithms, improving early competitive diagnosis with AUCs best performances at 90.55%.](#)
- [Conducted statistical and comparative analyses of Type 2 Diabetes Mellitus in Mexican patients, identifying significant clinical and anthropometric risk factors.](#)
- Presented research findings at international conferences, focusing on applications in anthropometric data analysis.
- Analyzed clinical data from Hospital Siglo XXI using RStudio to create detailed visualizations, revealing key gender-specific risk factors for diabetes.
- Published peer-reviewed papers on the integration of AI in biomedical applications.

Biomedical Expert at IMSS 33

Bahía de Banderas, Mexico | November 2021 – August 2023

- Directed and optimized preventive and corrective maintenance protocols, ensuring 95% equipment functionality covering 2500+ medical equipment.

- First aid on troubleshooting, issue detections and clinical staff training related to RIS/PACS on a router centralized architecture
- Corrective and preventive to reduce errors and gaps between technology and new staff entries
- Led technical teams in medical equipment and devices in imaging and intensive care unit areas.
- Developed training programs for a medical and technical staff to improve team proficiency
- Streamlined supplier communication, reducing equipment downtime.

Biomedical Management Manager at Clínica de Jesus

Chihuahua, Mexico | April 2021 – November 2021

- Managed a comprehensive maintenance schedule covering 50+ medical devices.
- Introduced predictive maintenance techniques, reducing unplanned downtime.
- Prepared detailed inventory reports, achieving 100% compliance with regulatory standards.
- Collaborated with cross-functional teams to address critical equipment needs.

Service Engineer at ITM Chihuahua

Monterrey, Mexico | July 2021

- Delivered high-quality preventive and corrective maintenance services for 30+ devices within tight deadlines.

IT Technician at Visteon Chihuahua

Chihuahua, Mexico | January 2020

- Conducted IT equipment inventory and maintenance, ensuring operational efficiency across multiple departments.

TECHNICAL SKILLS

- Machine Learning: Predictive model development, complex data analysis
- RIS/PACS management, corrective and preventive maintenance in a centralized router architecture
- SQL management
- Data Analysis: Data manipulation, statistical analysis, advanced visualization (Python, RStudio, LookerStudio)
- Statistical Analysis
- Clinical Research
- Git
- Programming Languages: Python (Pandas, TensorFlow, MLFlow, SHAP, NumPy, SciPy, Scikit-learn, Matplotlib, Plotly), RStudio, Looker and Tableau
- Medical Equipment Management: Preventive, predictive, and corrective maintenance, regulatory compliance

Languages

- Spanish: Native
- English: Advanced

Certifications

- Data Science using Machine Learning Algorithm with Big Data | DEC 2024
- Health informatics and marketpace | Healthcare Data Management and interfaces | by Illinois Tech | JAN 2026
- Introduction and preparation for AWS Certified Cloud Practitioner (CLF-C01) | FEB 2025

Key Achievements

- Published research on ovarian cancer AI detection methodologies in '*Network Modeling Analysis in Health Informatics and Bioinformatics*'.

- Presented studies on Type 2 Diabetes Mellitus at national conferences, contributing to public health knowledge.
- Developed predictive models improving early diagnosis accuracy by 15%.